

Algebraic Fractions

Name
Date
Course

Adding and Subtracting

$$\textcircled{1} \frac{x}{8} + \frac{3x}{7}$$

$$\frac{7x}{56} + \frac{24x}{56}$$

$$\boxed{\frac{31x}{56}}$$

$$\textcircled{2} \frac{5y}{6} - \frac{2y}{3}$$

$$\frac{5y}{6} - \frac{4y}{6}$$

$$\boxed{\frac{y}{6}}$$

$$\textcircled{3} \frac{z}{4} - \frac{5z}{6}$$

$$\frac{3z}{12} - \frac{10z}{12}$$

$$\boxed{\frac{-7z}{12}}$$

$$\textcircled{4} -\frac{3x}{8} + \frac{5x}{12}$$

$$-\frac{9x}{24} + \frac{10x}{24}$$

$$\boxed{\frac{x}{24}}$$

$$\textcircled{5} \frac{5y}{9} + (-\frac{y}{6})$$

$$\frac{10y}{18} + (-\frac{3y}{18})$$

$$\boxed{\frac{7y}{18}}$$

$$\textcircled{6} -\frac{2z}{5} + (-\frac{z}{6})$$

$$-\frac{12z}{30} + (-\frac{5z}{30})$$

$$\boxed{\frac{-17z}{30}}$$

$$\textcircled{7} -\frac{3a}{8} - \frac{5a}{6}$$

$$-\frac{9a}{24} - \frac{20a}{24}$$

$$\boxed{\frac{-29a}{24}}$$

$$\textcircled{8} 5b + \frac{2b}{3}$$

$$\frac{5b}{1} + \frac{2b}{3}$$

$$\frac{15b}{3} + \frac{2b}{3}$$

$$\boxed{\frac{17b}{3}}$$

$$\textcircled{9} \frac{3y}{7} - (-\frac{5y}{11})$$

$$\frac{3y}{7} + \frac{5y}{11}$$

$$\frac{33y}{77} + \frac{35y}{77}$$

$$\boxed{\frac{68y}{77}}$$

$$\textcircled{10} -\frac{z}{2} - (-\frac{5z}{6})$$

$$-\frac{z}{2} + \frac{5z}{6}$$

$$-\frac{3z}{6} + \frac{5z}{6}$$

$$\frac{2z}{6}$$

$$\boxed{\frac{z}{3}}$$

$$\textcircled{11} \frac{2x}{5} + \frac{4x}{9}$$

$$\frac{18x}{45} + \frac{20x}{45}$$

$$\boxed{\frac{38x}{45}}$$

$$\textcircled{12} \frac{3y}{10} - \frac{y}{5}$$

$$\frac{3y}{10} - \frac{2y}{10}$$

$$\boxed{\frac{y}{10}}$$

$$\textcircled{13} \frac{3a}{5} - \frac{5a}{6}$$

$$\frac{18a}{30} - \frac{25a}{30}$$

$$\boxed{\frac{-7a}{30}}$$

$$\textcircled{14} -\frac{3b}{4} + \frac{5b}{6}$$

$$-\frac{9b}{12} + \frac{10b}{12}$$

$$\boxed{\frac{b}{12}}$$

$$\textcircled{15} \frac{3c}{4} + (-\frac{c}{10})$$

$$\frac{15c}{20} + \frac{-2c}{20}$$

$$\boxed{\frac{13c}{20}}$$

$$\textcircled{16} -\frac{7x}{10} + (-\frac{5x}{8})$$

$$-\frac{28x}{40} + (-\frac{25x}{40})$$

$$\boxed{\frac{-53x}{40}}$$

$$\textcircled{17} -\frac{3a}{8} - \frac{5a}{7}$$

$$-\frac{21a}{56} - \frac{40a}{56}$$

$$\boxed{\frac{-61a}{56}}$$

$$\textcircled{18} \frac{2b}{5} - 2b$$

$$\frac{2b}{5} - \frac{2b}{1}$$

$$\frac{2b}{5} - \frac{10b}{5}$$

$$\boxed{\frac{-7b}{5}}$$

$$(19) \frac{2z}{7} - (-\frac{4z}{9})$$

$$\frac{2z}{7} + \frac{4z}{9}$$

$$\frac{18z}{63} + \frac{28z}{63}$$

$$\boxed{\frac{46z}{63}}$$

$$(20) -\frac{A}{6} - (-\frac{2A}{9})$$

$$-\frac{A}{6} + \frac{2A}{9}$$

$$-\frac{3A}{18} + \frac{4A}{18}$$

$$\boxed{\frac{A}{18}}$$

$$(21) \frac{3x}{10} + \frac{7x}{100}$$

$$\frac{30x}{100} + \frac{7x}{100}$$

$$\boxed{\frac{37x}{100}}$$

$$(22) \frac{5x}{44} + (-\frac{3x}{11}) + \frac{7x}{22}$$

$$\frac{5x}{44} + (-\frac{12x}{44}) + \frac{14x}{44}$$

$$-\frac{7x}{44} + \frac{14x}{44}$$

$$\boxed{\frac{7x}{44}}$$

$$(23) -\frac{5y}{7} + \frac{3y}{8} - (-\frac{11y}{56})$$

$$-\frac{5y}{7} + \frac{3y}{8} + \frac{11y}{56}$$

$$-\frac{40y}{56} + \frac{21y}{56} + \frac{11y}{56}$$

$$-\frac{8y}{56}$$

$$\boxed{-\frac{y}{7}}$$

$$(24) -\frac{2z}{3} - \frac{5z}{18} - \frac{7z}{9}$$

$$-\frac{12z}{18} - \frac{5z}{18} - \frac{14z}{18}$$

$$\boxed{-\frac{31z}{18}}$$

$$(25) \frac{4x}{5} - \frac{x}{6} + (-\frac{7x}{10})$$

$$\frac{24x}{30} - \frac{5x}{30} + (-\frac{21x}{30})$$

$$-\frac{2x}{30}$$

$$\boxed{-\frac{x}{15}}$$

$$(26) -\frac{5y}{8} - \frac{y}{4} + (-\frac{5y}{8})$$

$$-\frac{15y}{24} - \frac{6y}{24} + (-\frac{20y}{24})$$

$$\boxed{-\frac{41y}{24}}$$

$$(27) -4a - \frac{2a}{3} + \frac{a}{6}$$

$$-\frac{4a}{1} - \frac{2a}{3} + \frac{a}{6}$$

$$-\frac{24a}{6} - \frac{4a}{6} + \frac{a}{6}$$

$$-\frac{28a}{6} + \frac{a}{6}$$

$$-\frac{27a}{6} = \boxed{-\frac{9a}{2}}$$

$$(28) -3x + \frac{2x}{3} + \frac{x}{6}$$

$$-\frac{3x}{1} + \frac{2x}{3} + \frac{x}{6}$$

$$-\frac{18x}{6} + \frac{4x}{6} + \frac{x}{6}$$

$$\boxed{-\frac{13x}{6}}$$

$$(29) \frac{B}{6} + (-\frac{5B}{12}) + \frac{3B}{4}$$

$$\frac{2B}{12} + (-\frac{5B}{12}) + \frac{9B}{12}$$

$$\frac{6B}{12}$$

$$\boxed{\frac{B}{2}}$$

$$(30) -\frac{5c}{6} + \frac{3c}{5} - (-\frac{11c}{10})$$

$$-\frac{25c}{30} + \frac{18c}{30} + \frac{33c}{30}$$

$$\frac{26c}{30}$$

$$\boxed{\frac{13c}{15}}$$

$$(31) -\frac{7x}{8} - \frac{4x}{5} - \frac{9x}{20}$$

$$-\frac{35x}{40} - \frac{32x}{40} - \frac{18x}{40}$$

$$-\frac{85x}{40}$$

$$\boxed{-\frac{17x}{8}}$$

$$(32) \frac{3r}{4} + \frac{4r}{8} - r$$

$$\frac{6r}{8} + \frac{4r}{8} - \frac{r}{1}$$

$$\frac{6r}{8} + \frac{4r}{8} - \frac{8r}{8}$$

$$\frac{2r}{8}$$

$$\boxed{\frac{r}{4}}$$

$$(33) 2A - \frac{5A}{8} - (-\frac{A}{4})$$

$$\frac{2A}{1} - \frac{5A}{8} + \frac{A}{4}$$

$$\frac{16A}{8} - \frac{5A}{8} + \frac{2A}{8}$$

$$\boxed{\frac{13A}{8}}$$

$$(34) 3.2x - 5.6x = \boxed{-2.4x}$$

$$(39) 2x - 5.9x = \boxed{-3.9x}$$

$$(35) -12.9y + (-4y) = \boxed{-16.9y}$$

$$(40) -7.33x + (-4.8x) = \boxed{-12.13x}$$

$$(36) -1.42z - 5.8z = \boxed{-7.22z}$$

$$(41) -7.33x - 4.8x = \boxed{-12.13x}$$

$$(37) 6.2y + (-8.9y) = \boxed{-2.7y}$$

$$(42) -3.8y + 4.1y = \boxed{0.3y}$$

$$(38) -11.5z + (-5z) = \boxed{-16.5z}$$

$$(43) 97.4z + (-0.073z) = \boxed{97.327z}$$

Multiplying and Dividing

$$(44) \left(-\frac{3x^2}{7}\right) \frac{5x^8}{6} = -\frac{15x^{10}}{42} = -\frac{3 \cdot 5x^{10}}{6 \cdot 7} = -\frac{3 \cdot 5x^{10}}{2 \cdot 7} = -\frac{5x^{10}}{2 \cdot 7} = \boxed{-\frac{5x^{10}}{14}}$$

$$(45) \frac{2a^2}{3b^3} \left(-\frac{5a^4}{9b}\right) = \boxed{-\frac{10a^6}{27b^4}}$$

$$(46) -\frac{2c^2}{3d} \left(\frac{5d^4}{9c^3}\right) = -\frac{10c^2d^4}{27dc} = \boxed{-\frac{10d^3}{27c}}$$

$$(47) \left(-\frac{2x^2y^9}{7z^2}\right) \left(-\frac{5y^6z^6}{12x^8}\right) = \frac{2 \cdot 5x^2y^{15}z^6}{7 \cdot 12z^2x^8} = \frac{2 \cdot 5y^{15}z^4}{7 \cdot 2 \cdot 6x^6} = \boxed{\frac{5y^{15}z^4}{42x^6}}$$

$$(48) -\frac{5ab^3c^7}{7} \left(-\frac{b}{2c^5a^2}\right) = \frac{5ab^4c^2}{14c^5a^2} = \boxed{\frac{5b^4c^2}{14a}}$$

$$(49) 3e^2 \left(-\frac{2e^3}{5}\right) = \frac{3e^2}{1} \left(-\frac{2e^3}{5}\right) = \boxed{-\frac{6e^5}{5}}$$

$$(50) \frac{3}{5w^2} (-4w^7) = \frac{3}{5w^2} \left(-\frac{4w^7}{1}\right) = -\frac{12w^7}{5w^2} = \boxed{-\frac{12w^5}{5}}$$

$$(51) \frac{4x^2y^5}{5} \div \left(-\frac{3y^3}{8x^3}\right) = \frac{4x^2y^5}{5} \left(-\frac{8x^3}{3y^3}\right) = -\frac{32x^5y^2}{15y^3} = \boxed{-\frac{32x^5y^2}{15}}$$

$$(52) -\frac{3s^4r^4}{4h} \div \frac{9h^2r^2}{10s^6} = -\frac{3s^4r^4}{4h} \left(\frac{10s^6}{9h^2r^2}\right) = -\frac{5s^{10}r^2}{6h^3} = \boxed{-\frac{5s^{10}r^2}{6h^3}}$$

$$(53) -\frac{5a^7b^3}{11c^5} \div \left(-\frac{2b^4c^3}{9a^2}\right) = -\frac{5a^7b^3}{11c^5} \left(-\frac{9a^2}{2b^4c^3}\right) = \frac{45a^9b^3}{22b^4c^8} = \boxed{\frac{45a^9}{22c^8b}}$$

$$(54) \left(-\frac{7x^5}{8y}\right) \div \left(-\frac{9}{10xy}\right) = -\frac{7x^5}{8y} \left(-\frac{10xy}{9}\right) = \frac{35x^6y}{36y} = \boxed{\frac{35x^6}{36}}$$

$$(55) -7h^2g^7 \div \frac{3g^5}{4h^4} = -\frac{7h^2g^7}{1} \left(\frac{4h^4}{3g^5}\right) = -\frac{28h^6g^2}{3g^5} = \boxed{-\frac{28h^6g^2}{3}}$$

$$(56) \frac{5d^2}{6} \div (-3d^8) = \frac{5d^2}{6} \div \left(-\frac{3d^8}{1}\right) = \frac{5d^2}{6} \left(-\frac{1}{3d^8}\right) = \frac{5d^2}{-18d^8} = \boxed{-\frac{5}{18d^6}}$$

$$(57) \left(-\frac{2e^2}{5}\right) \frac{4e^4}{5} = \boxed{-\frac{8e^6}{25}}$$

$$(58) \frac{7a^2}{9b^3} \left(-\frac{5a^3}{8b}\right) = \boxed{-\frac{35a^5}{72b^4}}$$

$$(59) -\frac{3y^5}{4x} \left(\frac{5x^2}{7y^2}\right) = -\frac{15y^3x^2}{28xy^2} = \boxed{-\frac{15y^3x}{28}}$$

$$(60) \left(-\frac{2r^2}{9s^2h}\right) \left(-\frac{5s^6}{12r^4h^3}\right) = \frac{2 \cdot 5r^2s^6}{9 \cdot 12s^2r^4h^4} = \frac{2 \cdot 5s^4}{9 \cdot 2 \cdot 6r^2h^4} = \boxed{\frac{5s^4}{54r^2h^4}}$$

$$(61) -\frac{4a^3b^2}{7c^3a} \left(-\frac{b^5}{c^4a^2}\right) = \frac{4a^3b^7}{7c^7a^3} = \boxed{\frac{4b^7}{7c^7}}$$

$$(62) -\frac{e^2}{3} (4e^8) = -\frac{e^2}{3} \left(\frac{4e^8}{1}\right) = \boxed{-\frac{4e^{10}}{3}}$$

$$(63) (-4w^5) \frac{3}{5w^2} = \left(-\frac{4w^5}{1}\right) \frac{3}{5w^2} = -\frac{12w^5}{5w^2} = \boxed{-\frac{12w^3}{5}}$$

$$(64) \frac{x^2y^2}{5} \div \left(-\frac{3y^2}{8x^{11}}\right) = \frac{x^2y^2}{5} \left(-\frac{8x^{11}}{3y^2}\right) = -\frac{8x^{13}y^2}{15y^2} = \boxed{-\frac{8x^{13}}{15}}$$

$$(65) -\frac{3r^6g^9}{4s^2} \div \frac{9s^3g^2}{10r^6} = -\frac{3r^6g^9}{4s^2} \left(\frac{10r^6}{9s^3g^2}\right) = -\frac{5r^{12}g^7}{6s^5g^2} = \boxed{-\frac{5r^{12}g^7}{6s^5g^2}}$$

$$(66) -\frac{2ab^3}{c^5} \div \left(-\frac{b^5}{8a^3c^7}\right) = -\frac{2ab^3}{c^5} \left(-\frac{8a^3c^7}{b^5}\right) = \frac{16a^3b^3c^7}{c^5b^5} = \boxed{\frac{16a^3c^2}{b^2}}$$

$$(67) \left(-\frac{7k^5h^{12}}{8p^6}\right) \div \left(-\frac{9k}{10h^2p}\right) = -\frac{7k^5h^{12}}{8p^6} \left(-\frac{10h^2p}{9k}\right) = \frac{35k^5h^{14}p}{36p^6k} = \boxed{\frac{35k^4h^{14}}{36p^5}}$$

$$(68) \frac{4d^5}{7} \div (-2d^3) = \frac{4d^5}{7} \div \left(-\frac{2d^3}{1}\right) = \frac{4d^5}{7} \left(-\frac{1}{2d^3}\right) = -\frac{2d^5}{7d^3} = \boxed{-\frac{2d^2}{7}}$$

$$(69) -6r^3g^5 \div \frac{3g^7}{4r^2} = -\frac{6r^3g^5}{1} \left(\frac{4r^2}{3g^7}\right) = -\frac{8r^5g^5}{g^7} = \boxed{-\frac{8r^5}{g^2}}$$